Database Normalization – Cafeteria System

Phase 1: Unnormalized Form (0NF)
All data is stored in one large table:

Student ID	Name	Department	Date	Meal Session	Menu Details	Time
101	Abel	CS	2026-05-05	Breakfast	Bread,tea,rice	8:00 am

- Data redundancy (same student info repeated many times)
- Wastes storage
- Leads to inconsistency

Phase 2: First Normal Form (1NF)
Rule: Each field contains only atomic (single) values.
Application:
- Remove multi-valued attributes (e.g., "Rice, Stew, Salad")
- Each column holds a single value
Result:
- Tables have primary keys
- Data is structured properly

Student ID	Name	Department	Date	Meal Session	Menu Item	Time
101	Abel	CS	2026-05-05	Breakfast	Bread	8:00 am
101	Abel	CS	2026-05-05	Breakfast	tea	8:00 am
101	Abel	CS	2026-05-05	Breakfast	rice	8:00 am

Phase 3: Second Normal Form (2NF)

Composite key:

(StudentID, Date, MealSession)

Non-key attributes:

- StudentName
- Department

Problem:

- StudentName and Department depend only on StudentID, not the whole key.

Rule:
- Must be in 1NF
- No partial dependency (non-key attributes depend on whole primary key)

Application:
- StudentName and Department depend only on StudentID
Fix:
- Create separate STUDENT table
Result:
- Reduced redundancy
- Easier updates

STUDENT TABLE

studentID	studentName	Department

TRANSACTION Table

TransactionID	StudentID	Date	MealSession

Phase 4: Third Normal Form (3NF)
Rule:
- Must be in 2NF
- No transitive dependency (non-key depends on non-key)

MEAL_SESSION Table

SessionID	SessionName	StartTime

MENU Table

MenuID	MenuName

Phase 5: BCNF (Boyce-Codd Normal Form)
Rule:
- Every determinant must be a candidate key

Application:
- SessionID determines StartTime

Example:

If: SessionID → StartTime and SessionID is a candidate key, then the table satisfies BCNF.

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Result:
- All tables are fully normalized
- No redundancy remains

Summary:
1NF: Atomic values and primary keys
2NF: Removed partial dependencies
3NF: Removed transitive dependencies
BCNF: All determinants are candidate keys